



Automobiles of the future

As part of a future-gazing concept, a design consultancy IDEO has many interesting concepts on automobility <http://dgit.in/IDEODesign>



DSLR-like phone camera

Sony's new Exmor sensors could bring in DSLR-like autofocus in smartphone cameras <http://dgit.in/SonyExmor>

- bluetooth Usually not needed on a desktop, doubly not needed while gaming.
- cupsLinux's printing service.
- apache / nginx / mysql / postgresql Web and database servers that can sometimes be installed without your explicit knowledge when required by other software.
- samba This service allows networking with Windows users.

How do I stop the above services you ask? The answer depends on which Linux distro you use. For users of recent Ubuntu versions, you will need to run `sudo service <service_name> stop`

In most other recent distros the command you need to run is `sudo systemctl stop <service_name>`

If you use KDE software, also run `akonadictl stop` to turn off some services that run in the background syncing email, calendars etc. And `balooctl stop` to stop file indexing.

Turn off Desktop Composition

Desktop composition is the somewhat recent trend of using the graphics card to render the desktop environment, such as windows, window controls etc. It's generally a good thing since it utilises the GPU to do things it best designed for while freeing up the CPU. When playing a game though you GPU has better things to do.

On Gnome 3 and Unity, you cannot turn it off directly, but in recent versions the compositor recognises when a game is run and gives it all the resources (not a perfect solution, but it's the one you have). On KDE you can have it automatically suspended when there is a fullscreen window, or use the Alt-Shift-F12 shortcut. You can also type the following on the command line, or make it part of the script launching a game:

```
qdbus org.kde.kwin /KWin toggleCompositing
```

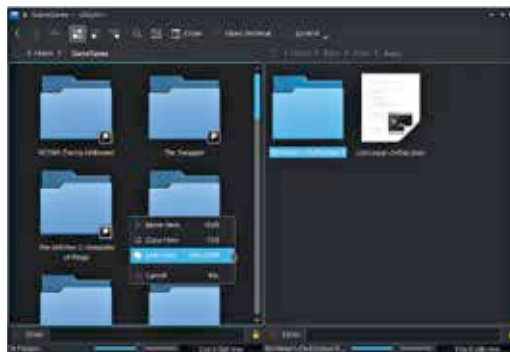
Close resource intensive applications

Perhaps this seems too obvious an advise to bear repeating. But we feel it important

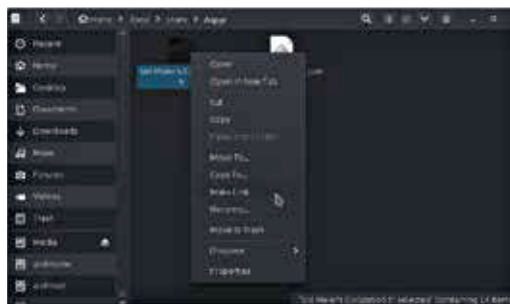
to note which kinds of application can be resource intensive.

For instance all browsers now endeavour to use the GPU to improve browsing performance, which makes them all the worse to have running in the background when you are playing a game. If the app uses the GPU at all (there will only be more such apps going on) then you should probably shut it down before you launch a game, and see if it reduces GPU usage.

Wait, but how does one know the GPU usage? Well, you can...



When you drag-drop a file in dolphin, you get the option to create a link.



Nautilus allows you to create a link in the same folder using the context menu.

Use performance monitoring tools for your GPU

Since such tools are tightly linked with the GPU driver, the tool you use will depend greatly on the graphics hardware on your system.

For instance if you have an NVIDIA card, you can use the handy `nvidia-smi` command line tool that comes with the official driver. You can run `nvidia-smi -l` to have it constantly update every few seconds. A graphical alternative would be `nvidia-settings`.

For AMD users, the `raedontop` command should work with both open- and

closed-source drivers. For Intel users there is `intel_gpu_top`. These tools might not be installed by default though.

Switch to a lighter desktop environment

This is a contentious recommendation so we feel that we need to clarify: this is not recommending a permanent switch. Keep using the desktop of your choice, just switch to a lighter one before a gaming session. After all, the fancy features of your favourite desktop environment are wasted when you are playing a game in full screen mode.

Openbox is a good choice, given that it is less than an MB in size. If you want to automatically start Steam when you launch OpenBox, you can add a `.desktop` file pointing to it in `~/.config/openbox/autostart`

You can even configure this to launch Steam directly in big picture mode by adding `-bigpicture` switch in the `.desktop` file.

If you have a Debian-based system like Ubuntu, you can install the SteamOS packages giving you almost all SteamOS features without needing to install it.

This could be a pain point since it will require you to log out of a running session, but know there are alternatives, just Google 'How to run multiple X sessions'. It solves many problems but is too involved to go into here.

Manage game saves better with symlinks

While both Windows and Linux have it this point of time an established standard for where games and other applications should store their data, most developers simply ignore the advice. The result is that save files are scattered all over the place.

The good news is, you can link all these into one single place, say a folder called 'Game Saves' in your home directory. Creating a link (in this case a symlink / symbolic link) on Linux is possible through both the GUI and command line. If you use Dolphin you can simply drag a folder / file from its original location to the destination, and you will see an option to move, copy or link the file. Choose to link it. In Nautilus (Gnome / Unity) you can right click any file / folder and choose 'Make Link' this will